

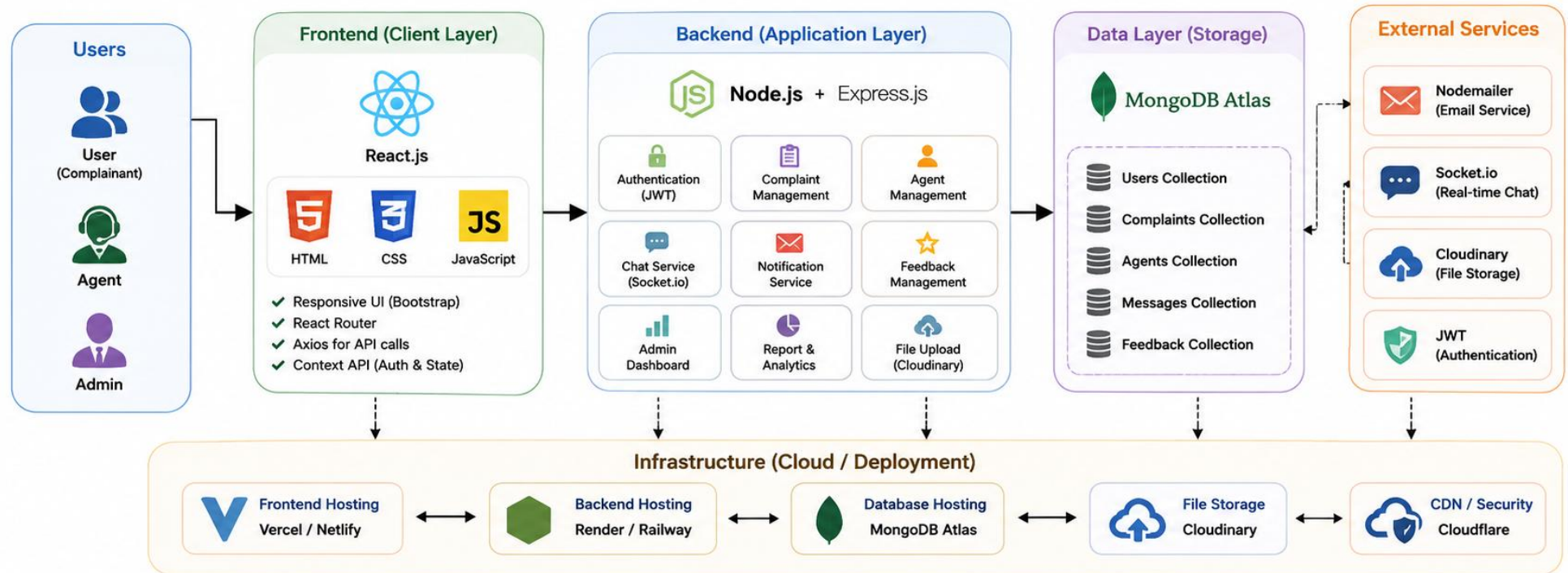
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	18 june 2026
Team ID	Individual project
Project Name	Online Complaint Registrations and Management System
Maximum Marks	4 Marks

Technical Architecture:

Technical Architecture

Online Complaint Registration and Management System (MERN Stack)



Flow Description:

1. Users interact with the system via React.js frontend.
2. Frontend communicates with backend APIs using Axios.
3. Backend processes requests and interacts with MongoDB.
4. External services handle email, real-time chat, and file storage.



Security:

- JWT Authentication
- Password Encryption (bcryptjs)
- Role Based Access Control
- Helmet & CORS Protection



Key Features:

- Complaint Registration & Tracking
- Agent Assignment & Resolution
- Real-time Chat
- Email Notifications
- Reports & Analytics



Tech Stack (MERN):

- MongoDB Atlas
- Express.js
- React.js
- Node.js

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web application for Users, Agents and Admin	HTML, CSS, JavaScript, React.js, Bootstrap
2	Application Logic-1	User Authentication & Authorization	Node.js, Express.js, JWT, bcryptjs
3	Application Logic-2	Complaint Registration & Tracking	Node.js, Express.js
4	Application Logic-3	Agent Assignment & Complaint Resolution	Node.js, Express.js
5	Database	Store users, complaints, agents, feedback	MongoDB, Mongoose
6	Cloud Database	Cloud-hosted database	MongoDB Atlas
7	File Storage	Complaint attachments and documents	Local Storage / Cloudinary
8	External API-1	Email Notifications	Nodemailer SMTP
9	External API-2	Real-time Communication	Socket.io
10	Machine Learning Model	Not Applicable	N/A
11	Infrastructure (Server/Cloud)	Application Deployment	Render, Vercel, MongoDB Atlas

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used in development	React.js, Node.js, Express.js, MongoDB
2	Security Implementations	Authentication, encryption, access control	JWT, bcryptjs, Helmet, CORS, Express Validator
3	Scalable Architecture	Three-tier architecture with separate frontend, backend, database	MERN Stack Architecture
4	Availability	Cloud deployment with high uptime	Render, Vercel, MongoDB Atlas
5	Performance	Fast API response and optimized database queries	Axios, MongoDB Indexing, React Hooks, Mongoose